CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The Mentor Track system is a Salesforce-based academic monitoring platform designed to help mentors track student performance and provide timely academic support. The system stores student personal details, academic records, and co-curricular activities, enabling mentors to monitor student progress effectively. The system automatically analyzes academic performance based on parameters such as attendance, internal assessment marks, CGPA, and end-semester marks. When a student's performance falls below predefined thresholds, the system generates an automated alert to the mentor, allowing them to identify students who require academic support.

Mentors can review the student's profile, analyze weak subjects, and assign subject-specific improvement tasks such as K-study sessions, seminar presentations, lab exercises, assignments, or mini-projects. These tasks are stored in the system and assigned through a structured workflow. Students receive automated email notifications with details of the assigned tasks, ensuring they are aware of the required academic interventions. The system also tracks task assignments and student responses, allowing mentors to monitor progress and evaluate improvement. In addition to academic monitoring, the system records co-curricular activities such as sports, cultural events, and inter-college competitions, providing mentors with a holistic view of student engagement and development.

The system also includes reports and dashboards that display academic trends, low-performing students, and extracurricular participation. These analytics help mentors quickly identify areas requiring attention and support better academic decision-making. Overall, the Mentor Track system improves the mentor-mentee interaction process by automating performance monitoring, providing early academic support, and enabling structured intervention strategies to enhance student success Furthermore, the system ensures timely communication between mentors and students, improving guidance and support. It also enhances transparency and accountability in tracking student progress, leading to better academic outcomes.

1.2 EXISTING SYSTEM

The Smart Student Academic Progress Monitoring and Early Risk Detection System is designed to help educational institutions effectively monitor and evaluate students' academic performance. In many institutions, tracking the progress of students is often done manually, which can be time-consuming and may lead to delays in identifying students who are struggling academically. This system provides a smart and automated solution that allows faculty members and mentors to continuously monitor students' academic activities such as attendance, internal assessment marks, assignment submissions, and overall performance. By maintaining all student academic records in a centralized digital platform, the system makes it easier for faculty to access and analyze student data efficiently.

The system also focuses on identifying students who are at academic risk at an early stage. Using predefined performance criteria such as low marks, poor attendance, or declining academic performance, the system can automatically classify students into different performance categories such as high performing, average, and high-risk students. Once a student is identified as high risk, the system can generate alerts or notifications to mentors and faculty members so that they can provide timely academic guidance and support. This early intervention helps prevent academic failure and encourages students to improve their performance.

In addition to monitoring performance, the system also improves communication between faculty, mentors, and students. Mentors can receive detailed reports about students who require attention and can take appropriate actions such as counseling, extra academic support, or follow-up monitoring. The system may also generate reports in formats such as dashboards, charts, or downloadable documents to help faculty easily understand students' academic trends and progress over time. Overall, the Smart Student Academic Progress Monitoring and Early Risk Detection System enhances the efficiency of academic monitoring and supports better decision-making in educational institutions.

Providing real-time insights into student performance and enabling early identification of academic risks, the system helps institutions improve student success rates and ensure that every student receives the necessary support to achieve their academic goals. It enables mentors to take proactive measures through timely interventions and personalized guidance. The system also enhances data-driven decision-making by offering detailed reports and analytics.

1.3 DRAWBACKS OF EXISTING SYSTEM

The main drawbacks are:

- Manual maintenance of student records such as attendance and marks.
- Lack of a centralized system to store and manage student academic data.
- Difficulty in tracking students' academic progress regularly.
- Delay in identifying students who are at academic risk.
- Limited communication between faculty members and mentors.
- Time-consuming process for generating student performance reports.
- Higher chances of human errors in data entry and record maintenance.
- No automated alerts or notifications for low-performing students.
- Lack of proper analysis of student academic performance.
- Inefficient monitoring of overall student progress.

1.4 PROPOSED SYSTEM

The proposed system is the Smart Student Academic Progress Monitoring and Early Risk Detection System, which is designed to overcome the limitations of the existing manual process used to track student academic performance. This system provides a digital platform where faculty members can easily record, monitor, and analyze students' academic details such as attendance, internal marks, and overall performance. In this system, faculty members can enter and manage student academic information through an online system. The system continuously analyzes the academic data to identify students who may be at academic risk based on predefined criteria such as low marks, poor attendance, or declining performance. Once a student is identified as a high-risk student, the system automatically generates alerts and sends notifications to the assigned mentor.

The mentor can view the list of high-risk students along with their academic details such as roll number, department, attendance, and marks. The system can also generate reports of these students and send them through email in formats such as PDF so that mentors can take necessary actions like counseling, academic support, or monitoring their improvement. The proposed system is implemented using the Salesforce Developer Platform, which provides role-based access for different users such as faculty members and mentors. Faculty members can update student academic records, while mentors can monitor student performance and receive risk alerts.

1.5 ADVANTAGES OF PROPOSED SYSTEM

The proposed Smart Student Academic Progress Monitoring and Early Risk Detection System provides several advantages compared to the existing manual system used to track student academic performance. The major advantages of the proposed system are listed below

- **Automated Academic Monitoring:** The system automatically records and monitors student academic details such as attendance, internal marks, and overall performance.
- **Early Identification of At-Risk Students:** The system analyzes student performance and identifies students who are at academic risk based on predefined criteria like low marks or poor attendance.
- **Mentor Alert System:** When a student is identified as high risk, the system sends alerts or notifications to the assigned mentor for immediate attention.
- **Centralized Student Data Management:** All student academic information is stored in a single system, making it easier to access, update, and manage records.
- **Improved Decision Making:** Faculty and mentors can easily analyze student performance through reports and dashboards to make better academic decisions.
- **Reduced Manual Work:** Automation reduces the need for manual record maintenance and minimizes administrative workload.
- **Better Communication Between Faculty and Mentors:** Mentors receive timely updates about students who need guidance or support.
- **Accurate and Reliable Data:** The system reduces human errors in data entry and ensures accurate academic records.
- **Efficient Performance Tracking:** Faculty members can continuously track student progress across subjects and semesters.
- **Improved Student Support:** Early detection of academic risks allows mentors and faculty to provide timely guidance and help students improve their performance.
- **Enhanced Transparency:** All academic activities and student progress are clearly recorded, ensuring transparency in monitoring and evaluation.
- **Secure Data Management:** The system ensures data security by restricting access based on user roles such as faculty and mentor.

In The proposed system enhances the overall efficiency of academic management by providing a structured and systematic approach to monitoring student performance. It reduces delays in identifying academic issues and enables timely intervention by mentors and faculty members. The system also supports continuous evaluation by maintaining updated records of student progress, which helps in long-term academic planning. By integrating multiple features such as performance tracking, reporting, and alert mechanisms, the system creates a comprehensive platform that improves both teaching effectiveness and student learning outcomes.

In addition, the system promotes a more organized and structured academic environment by digitizing all student-related information. It eliminates the need for maintaining physical records and reduces the chances of data loss or duplication. Faculty members can easily update and retrieve information at any time, which improves efficiency and saves time. The availability of centralized data also helps in maintaining consistency across different departments.

Moreover, the system supports continuous improvement in academic performance by enabling regular monitoring and evaluation. It encourages proactive mentoring by allowing mentors to track student progress and provide timely feedback. This helps students stay focused on their academic goals and improves their overall performance. The system also creates a supportive learning environment where both students and mentors can collaborate effectively for better outcomes.

SUMMARY

This chapter introduced the Smart Student Academic Progress Monitoring and Early Risk Detection System and highlighted the need for an automated system to monitor student academic performance. It discussed the drawbacks of the existing manual system such as difficulty in tracking student progress, delayed identification of at-risk students, and lack of centralized data management. The proposed system uses the Salesforce platform to record and analyze student academic data, identify high-risk students, and notify mentors for timely support. This system improves efficiency, accuracy, and overall academic monitoring within the institution. It also enhances transparency in academic evaluation and ensures that all student data is easily accessible for analysis. The system plays a key role in improving communication between faculty and mentors, leading to better student guidance and outcomes.

CHAPTER 2

SYSTEM ANALYSIS

2.1 IDENTIFICATION ON NEED

In many educational institutions, monitoring student academic performance is still carried out using manual methods such as paper records or basic spreadsheets. These traditional methods make it difficult for faculty members to maintain and analyze large amounts of student academic data efficiently. Managing information such as attendance, internal marks, and overall performance becomes time-consuming and increases the chances of errors and data loss. Faculty members often face difficulties in tracking the academic progress of students continuously. It also becomes challenging to identify students who are at academic risk at an early stage.

In many cases, students who are struggling academically are identified only after examination results are published, which may be too late to provide effective support or corrective measures. Additionally, the lack of proper communication between faculty members and mentors makes it difficult to provide timely guidance to students who require academic assistance.

Therefore, there is a clear need for an automated system that can effectively monitor and manage student academic performance. The Smart Student Academic Progress Monitoring and Early Risk Detection System is designed to address these issues by providing a centralized digital platform where faculty members can record and analyze student academic data. The system helps identify students who are at academic risk and automatically notifies mentors so that timely guidance and support can be provided. This system improves transparency, reduces manual work, and ensures better academic monitoring within the institution.

2.2 FEASIBILITY STUDY

A feasibility study is an important stage in system development that helps determine whether a proposed system is practical and beneficial for implementation. It evaluates different factors such as available resources, technology, cost, and time required to successfully develop and operate the system. Conducting a feasibility study helps ensure that the system can be implemented effectively without causing unnecessary difficulties or excessive costs.

In the case of the Smart Student Academic Progress Monitoring and Early Risk Detection System, the feasibility study is conducted to analyze whether the proposed system can effectively address the problems faced in monitoring student academic performance using traditional methods. In many institutions, tracking student progress manually can be time-consuming and may lead to delays in identifying students who are at academic risk. The proposed system aims to provide an automated platform where students can enter their academic details, attendance, and co-curricular activities, while mentors can monitor performance and assign improvement tasks when necessary.

The feasibility study also evaluates whether the system can be implemented using the available technological resources, such as the Salesforce platform, and whether it can be developed within the required time and cost constraints. Since the system uses a cloud-based platform, it reduces the need for complex infrastructure and allows easy data management and accessibility. It helps educational institutions identify academically at-risk students at an early stage and provide timely guidance through mentors, thereby improving overall student performance and academic support.

2.2.1 Economic Feasibility

Economic feasibility evaluates whether the proposed system is cost-effective compared to the existing methods of monitoring student academic performance. The Smart Student Academic Progress Monitoring and Early Risk Detection System is developed using the Salesforce platform, which reduces the need for additional hardware or infrastructure. The system mainly requires internet access and basic computing devices that are already available in most educational institutions. By automating the process of recording academic performance, attendance, and co-curricular activities, the system reduces manual work, paperwork, and the effort required for maintaining student records.

As a result, the cost involved in implementing and maintaining the system is minimal while providing significant benefits such as improved efficiency, early identification of academically at-risk students, and better management of student performance data. In addition, the system reduces operational costs by minimizing human effort and administrative overhead. It also saves time for faculty members, allowing them to focus more on teaching and mentoring activities. The scalability of the system ensures that it can be expanded without significant additional costs, making it a sustainable and economical solution for educational institutions.

2.2.2 Operational Feasibility

Operational feasibility determines whether the proposed system can be effectively used by the users involved in the institution. The Smart Student Academic Progress Monitoring and Early Risk Detection System is designed with a simple and user-friendly interface that allows both students and mentors to easily interact with the system. Students can enter their personal details, academic marks, attendance information, and co-curricular activities through the portal. The system analyzes the academic data and identifies students who are performing poorly. When a student is identified as a high-risk student, an alert is automatically sent to the mentor. The mentor can then review the student's performance and assign tasks or guidance to help improve their academic progress. Since the system simplifies academic monitoring and clearly defines the roles of students and mentors, it can be easily adopted and effectively used within educational institutions.

2.2.3 Technical Feasibility

Technical feasibility examines whether the required technology and technical resources are available to develop and maintain the system. The proposed system is implemented using the Salesforce Experience Cloud platform, which provides a secure and reliable environment for developing web-based applications. The platform supports features such as data management, automation using flows, user authentication, and role-based access for different users such as students and mentors. Additionally, Salesforce provides built-in tools for creating objects, reports, dashboards, and automated notifications, which makes system development easier and more efficient. Since the platform operates on cloud infrastructure, it eliminates the need for complex hardware installations and allows easy access to the system from any location with internet connectivity. Therefore, the proposed system is technically feasible and can be successfully implemented using the available technology.

2.3 SOFTWARE REQUIREMENT SPECIFICATIONS

Software Requirement Specifications (SRS) describe the software and hardware requirements needed to develop and run the proposed system. It provides a clear understanding of the tools, technologies, and resources required for the successful implementation of the Smart Student Academic Progress Monitoring and Early Risk Detection System.

2.3.1 Software Requirements

The software requirements specify the applications and technologies needed to develop and operate the system. The main software used for developing the Smart Student Academic Progress Monitoring and Early Risk Detection System includes:

- Operating System: Windows 10 or above
- Platform: Salesforce Platform
- Development Environment: Salesforce Developer Edition
- Technology Used: Salesforce Experience Cloud, Salesforce Flows, Email Alerts, Reports and Dashboards
- Browser: Google Chrome / Microsoft Edge
- Documentation Tool: Microsoft Word

These software tools help in designing, developing, and managing the system efficiently. The Salesforce platform provides a cloud-based environment that supports data storage, automation, and user management. The system also uses Email Alert functionality to notify mentors when a student is identified as academically at risk, allowing mentors to take timely action and assign tasks to improve student performance.

2.3.2 Hardware Requirements

Hardware requirements refer to the physical components needed to run the system efficiently. The Smart Student Academic Progress Monitoring and Early Risk Detection System is a web-based application developed using the Salesforce platform, so it does not require high-end hardware. The system can run on standard computers available in most educational institutions as long as they have internet connectivity.

- Processor: Intel Core i3 or above
- RAM: 4 GB or higher
- Hard Disk: 250 GB or higher
- Display: Standard monitor with internet connectivity
- Input Devices: Keyboard and Mouse
- Network: Stable internet connection for accessing the Salesforce cloud platform

These hardware components are sufficient to access the system through a web browser and allow students and mentors to update academic data, monitor performance, and manage mentoring tasks effectively.

2.4 SOFTWARE DESCRIPTION

2.4.1 Front End

The front end of the Smart Student Academic Progress Monitoring and Early Risk Detection System is developed using Salesforce Experience Cloud. It provides a web-based user interface that allows users to interact with the system through an online portal. The front end acts as the communication layer between the users and the system, enabling them to access system features easily through a web browser. Through the portal, students can log in to the system and enter their personal details, academic performance information such as marks and attendance, and co-curricular activities.

The system stores this information and analyzes it to identify students who may be academically at risk. If a student's performance falls below the predefined criteria, the system automatically triggers an email alert to the assigned mentor. Mentors can log in to the portal to view student academic reports, monitor the performance of their assigned students, and receive notifications about high-risk students. Based on the analysis, mentors can assign tasks or provide guidance to help students improve their academic performance.

Students can also view the tasks assigned by mentors and respond through the system. Salesforce Experience Cloud provides a secure, responsive, and user-friendly interface that allows both students and mentors to access the system from any location with internet connectivity. The platform also supports features such as navigation menus, record forms, dashboards, and reports, which help users easily manage and view academic information. Overall, the front end of the system ensures efficient interaction between students and mentors, simplifies academic monitoring, and improves the overall usability of the platform, making it easier for institutions to track student progress and provide timely academic support.

2.4.2 Back End

The back end of the Smart Student Academic Progress Monitoring and Early Risk Detection System is developed using the Salesforce Platform. The back end is responsible for managing the system logic, data storage, automation processes, and communication between different system components. It ensures that the data entered by users is stored securely and processed correctly.

The system uses Salesforce objects and database features to store important information such as student personal details, academic performance records, attendance data, co-curricular activities, mentor details, and assigned tasks. These records are managed using the Salesforce cloud database, which provides secure and reliable data storage. Automation in the system is implemented using Salesforce Flows and Email Alerts. When students enter their academic details such as marks and attendance, the system automatically analyzes the data to detect students who may be academically at risk. If the system identifies low marks or poor attendance, an email alert is automatically sent to the assigned mentor. This helps mentors quickly identify students who need academic support.

Mentors can then review the student's performance and assign intervention tasks or guidance through the system. These tasks are stored in the database and displayed in the student portal so that students can view and respond to them. The back end also supports features such as reports and dashboards, which help mentors monitor student progress and analyze academic performance trends. Since the system is built on the Salesforce cloud platform, it provides high security, scalability, and reliability. The cloud-based architecture eliminates the need for complex hardware infrastructure and allows users to access the system from anywhere using an internet connection. This ensures efficient data management and smooth operation of the academic monitoring system.

2.4.3 Database

The database used in the Smart Student Academic Progress Monitoring and Early Risk Detection System is the Salesforce Cloud Database. It is a cloud-based database that stores all the information related to student details, academic performance, attendance records, co-curricular activities, mentoring tasks, and performance analysis. The Salesforce Cloud Database allows data to be stored, managed, and accessed securely through the Salesforce platform. It supports real-time data updates and ensures that all student-related information is organized and easily retrievable. The database uses custom objects and fields to store and manage the data required for monitoring student academic performance and identifying students who may be at academic risk. In this system, different objects are used to manage various types of information. For example, student information is maintained in the Student object, while academic marks and attendance are recorded in the Academic Performance object. Co-curricular activities are stored in a separate object to track student participation.

Since the database is cloud-based, it provides several advantages such as secure data storage, automatic backup, data consistency, and easy access from different locations through the internet. Authorized users such as students and mentors can access the required information through the portal while maintaining proper data security and integrity. This ensures efficient management of academic records and supports the system in identifying and assisting students who require academic improvement.

In addition, the Salesforce database supports role-based access control, ensuring that only authorized users can view or modify specific data. It also provides high scalability, allowing the system to handle an increasing number of student records without performance issues. The use of validation rules and data constraints ensures data accuracy and prevents incorrect data entry. Reports and dashboards are directly generated from the database, enabling real-time analysis of student performance. Overall, the database plays a crucial role in maintaining data reliability, security, and efficient system operation.

Salesforce database architecture supports seamless integration with other modules, enabling smooth data flow across the system. Cloud-based storage ensures high availability and minimal downtime, improving system reliability. Advanced security features such as encryption and authentication help protect sensitive student information from unauthorized access. Data can be easily updated and synchronized across different users in real time, ensuring consistency. Efficient query and retrieval mechanisms allow quick access to large volumes of data, enhancing system performance.

SUMMARY

This chapter explained the system analysis of the Smart Student Academic Progress Monitoring and Early Risk Detection System. It discussed the need for the system and the feasibility study conducted to determine whether the proposed system is practical and beneficial for monitoring student academic performance. The chapter also described the software and hardware requirements necessary for developing and running the system. In addition, the system architecture was explained by identifying the front end, back end, and database components used in the application. The front end is developed using Salesforce Experience Cloud, which provides a user-friendly portal for students and mentors. The back end is implemented using the Salesforce platform, which manages data processing, automation, and email alerts for identifying academically at-risk students. The system database uses the Salesforce Cloud Database, which securely stores student information, academic records, and mentoring tasks.

CHAPTER 3

SYSTEM DESIGN

3.1 MODULE DESCRIPTION

The Smart Student Academic Progress Monitoring and Early Risk Detection System is divided into several modules to ensure proper organization and efficient functioning of the system. Each module performs specific tasks and manages different activities related to student academic monitoring and mentoring. The system helps track student performance, identify academically at-risk students, and allow mentors to provide guidance through intervention tasks.

The following modules are included in the system:

- Student Module
- Mentor Module
- Academic Performance Module
- Co-Curricular Activities Module
- Subject Intervention Task Module
- Mentee Response Module
- Reports and Dashboard Module

3.1.1 Student Module:

The Student Module is designed for students to enter and manage their academic information through the portal. In this module, students can provide their personal details, academic performance information such as marks and attendance, and co-curricular activities. This information is stored in the system and used to monitor the academic progress of each student. The system analyzes the student's academic data to identify students who may be academically at risk. If the student has low marks or poor attendance, the system automatically triggers an email alert to the assigned mentor. Students can also log in to the portal to view the intervention tasks assigned by their mentor. These tasks are designed to help students improve their academic performance in specific subjects. After completing the assigned tasks, students can submit their responses or progress updates through the system.

3.1.2 Mentor Module

The Mentor Module is responsible for monitoring the academic performance of students and providing guidance when necessary. In this module, mentors can view the academic details of the students assigned to them, including marks, attendance records, and other academic information entered by the students. The system analyzes the academic data and identifies students who may be academically at risk due to low marks or poor attendance. When such a situation occurs, the system automatically sends an email alert to the mentor, informing them about the student's academic status.

After receiving the alert, the mentor can log in to the portal and review the student's academic performance and progress. Based on this analysis, the mentor can assign intervention tasks or academic guidance to help the student improve their performance in specific subjects. This module helps mentors effectively monitor student progress, provide timely support, and ensure that students receive proper guidance to overcome academic difficulties and improve their overall performance.

3.1.3 Academic Performance Module

The Academic Performance Module is responsible for recording and monitoring the academic performance of students. In this module, students can enter their academic details such as subject name, marks obtained, and attendance percentage. These details are stored in the system and used to analyze the academic progress of each student. The system automatically evaluates the academic data entered by the student to identify students who may be academically at risk.

If the system detects low marks or poor attendance, it triggers an alert and sends an email notification to the assigned mentor. This helps mentors identify students who need additional academic support. This module plays an important role in continuously tracking student academic performance and ensuring that students receive timely guidance to improve their academic results.

3.1.4 Co-Curricular Activities Module

The Co-Curricular Activities Module allows students to record and manage information related to their extracurricular activities. In this module, students can enter details such as activity name, type of activity, achievements, and participation dates. These activities help provide a broader understanding of the student's overall development beyond academic performance.

The information stored in this module can also be viewed by mentors to understand the student's engagement in extracurricular activities. This module helps maintain a complete profile of the student and supports a more comprehensive evaluation of student progress. The information stored in this module can also be viewed by mentors to understand the student's engagement in extracurricular activities. This module helps maintain a complete profile of the student and supports a more comprehensive evaluation of student progress. It also enables mentors to identify students with strong interests and talents in non-academic areas.

3.1.5 Subject Intervention Task Module

The Subject Intervention Task Module allows mentors to assign improvement tasks to students who are identified as academically at risk. After receiving an alert from the system, the mentor can review the student's academic performance and assign specific tasks related to the subject in which the student is facing difficulties. The mentor can provide details such as task name, description, subject, and type of intervention. The assigned tasks are stored in the system and made visible to the student through the portal. This module helps mentors guide students by providing structured academic support to improve their performance in weak subjects.

3.1.6 Mentee Response Module

The Mentee Response Module allows students to respond to the tasks assigned by mentors. After completing the assigned task, students can submit their response or update their progress through the system. Students can provide details about the work completed, solutions prepared, or improvements made in the assigned subject. These responses are stored in the system and can be reviewed by mentors to evaluate the student's progress. This module encourages active participation from students and allows mentors to track whether the assigned interventions are helping students improve their academic performance.

The module also supports continuous monitoring by maintaining a history of student responses, which helps mentors analyze progress over time. Students can update their responses multiple times if needed, allowing them to reflect improvements and corrections. Status indicators such as completed, pending, or in progress can be used to track task completion effectively. The module promotes accountability among students and ensures that assigned tasks are completed within the given timeframe.

3.1.7 Report and Dashboard Module

The Report and Dashboard Module helps mentors and administrators analyze and monitor student academic data effectively. This module provides summarized information about student performance, risk levels, and mentoring activities. Through reports, users can view details such as students with low marks, students with poor attendance, tasks assigned by mentors, and responses submitted by students. The dashboard presents this information in the form of charts and graphs, making it easier to understand student performance trends. This module helps mentors and administrators monitor student progress, identify academically at-risk students quickly, and make better academic decisions based on real-time data.

3.1.8 Admin Module

The Admin Module is responsible for managing and controlling the overall functionality of the Smart Student Academic Progress Monitoring and Early Risk Detection System. The administrator manages user accounts, system settings, and ensures that the system operates smoothly. Through this module, the administrator can create and manage user roles such as students and mentors, configure system settings, and monitor system activities. The admin is also responsible for maintaining data security and ensuring that only authorized users can access the system. This module plays an important role in maintaining system stability, controlling user access, and ensuring that the academic monitoring process functions efficiently.

3.2 DATA FLOW DIAGRAM

A Data Flow Diagram is a graphical representation that shows the flow of data through a system. It illustrates how data moves from one entity to another, how it is processed, and how it is stored. DFDs help in understanding the overall system functionality and interactions between external entities and internal processes.

3.2.1 Data Flow Diagram Level 0

Data Flow Diagram Level 0 provides a high-level overview of the system's data flow by representing the system as a single process and showing interactions with external entities. It focuses on major data exchanges without detailing internal processing steps.

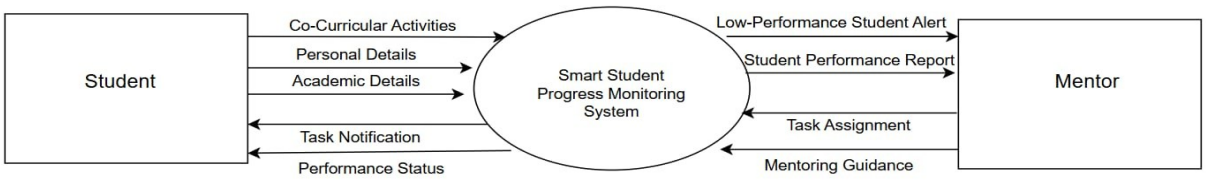

Fig 3.1 Data Flow Diagram Level 0

The Data Flow Diagram Level-0 represents the Smart Student Academic Progress Monitoring and Early Risk Detection System as a single central process that interacts with the external entities involved in monitoring student academic performance. In this diagram, the Student is one of the primary users who interacts with the system by entering important information such as personal details, academic performance details, and co-curricular activities. These details are submitted to the system and stored for further analysis. The system processes the academic information and evaluates the student's performance based on factors such as marks and attendance.

After analyzing the student data, the system generates a student performance report. If the system detects that a student has low marks or poor attendance, it automatically generates a low-performance student alert and sends it to the mentor through an email notification. This alert helps mentors quickly identify students who require academic support. The Mentor receives the performance report and reviews the student's academic progress. Based on this analysis, the mentor can provide mentoring guidance and assign intervention tasks to help the student improve their performance.

The assigned task is sent to the system, which then delivers a task notification to the student. The student can view the task and work on the assigned activity to improve their academic performance. The system also provides performance status updates to the student so they can monitor their academic progress. Thus, the Smart Student Academic Progress Monitoring System acts as the central processing unit that collects student data, analyzes academic performance, identifies at-risk students, sends alerts to mentors, and supports mentoring activities to improve student performance. The Level-0 Data Flow Diagram therefore highlights the interaction between students, mentors, and the monitoring system. This improves the efficiency of the overall academic monitoring process and ensures timely support for students who require guidance.

3.2.2 Data Flow Diagram Level 1

The Data Flow Diagram Level-1 diagram represents the detailed working process of the Smart Student Academic Progress Monitoring and Early Risk Detection System. It explains how student data is collected, analyzed, and used to support mentoring activities.

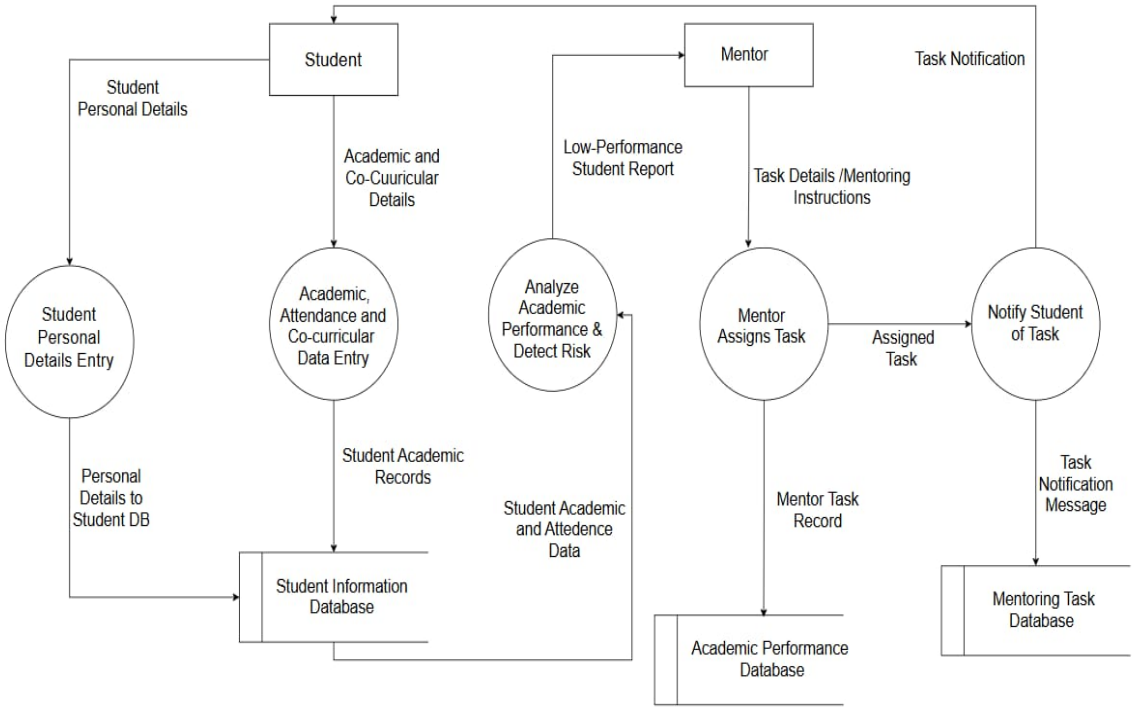

Figure 3.2 Data Flow Diagram Level 1

In Figures 3.2, the student first enters their personal details through the Student Personal Details Entry process. These details are stored in the Student Information Database, which maintains basic information about each student. Next, the student enters their academic performance, attendance information, and co-curricular activities through the Academic, Attendance and Co-curricular Data Entry process. This information is also stored in the Student Information Database. The stored data is later used for monitoring the academic progress of students. The system then processes the stored academic data through the Analyze Academic Performance and Detect Risk process.

In this stage, the system evaluates student marks and attendance records to identify students who may be academically at risk. If the system detects low marks or poor attendance, it generates a low-performance student report and sends it to the mentor. The mentor reviews the report and provides appropriate guidance. Through the Mentor Assigns Task process, the mentor can assign intervention tasks or mentoring instructions to help the student improve their academic performance. The assigned tasks are recorded in the Academic Performance Database. After the task is assigned, the system processes the information through the Notify Student of Task process.

The student receives a task notification message informing them about the mentoring task assigned by the mentor. These notifications are stored in the Mentoring Task Database for record keeping. Thus, the DFD Level-1 diagram shows how student information flows through different processes of the system, how academic performance is analyzed, and how mentors provide guidance through assigned tasks. It highlights the interaction between students, mentors, system processes, and databases to ensure effective monitoring and improvement of student academic performance.

3.3 SYSTEM FLOW DIAGRAM

The System Flow Diagram illustrates the step-by-step process involved in the Smart Student Academic Progress Monitoring and Early Risk Detection System. It explains how student information is collected, analyzed, and used to support mentoring activities for improving academic performance. The process begins when the student logs into the system using their credentials. After logging in, the student enters their personal details, which are stored in the system for identification and record maintenance.

Next, the student provides academic details such as subject marks and attendance information. The student can also enter information about their co-curricular activities, which helps in maintaining a complete student profile. After all the information is entered, the system stores the student data in the database. The stored data is then processed through the performance analysis stage, where the system evaluates the student's academic performance based on marks and attendance. The system then sends a task notification to the student, informing them about the assigned mentoring task. The student completes the assigned task and submits their response through the system. Finally, the process ends after the student submits the response.

Figure 3.3 System Flow Diagram

In Figure 3.3, the system checks whether the student has low marks or low attendance. If the student's performance is normal, the process ends and the student can continue their academic activities without intervention.

However, if the system detects that the student is academically at risk, an email alert is automatically sent to the mentor. The mentor then reviews the student performance report and analyzes the areas where the student needs improvement. Based on this analysis, the mentor assigns an intervention task or guidance activity to help the student improve their academic performance.

3.4 INPUT DESIGN

Input design is an important part of system design that focuses on how data is entered into the system. It ensures that the data collected from users is accurate, complete, and easy to process. Proper input design helps reduce errors and improves the overall efficiency of the system. In the Smart Student Academic Progress Monitoring and Early Risk Detection System, input data is collected from students and mentors through a user-friendly web interface developed using the Salesforce platform. The system provides structured forms that allow users to enter the required information such as personal details, academic performance, attendance records, co-curricular activities, and mentoring tasks. The input forms are designed to ensure that only valid and necessary information is entered into the system. Various validation techniques such as required fields, proper data formats, and dropdown selections are used to prevent incorrect data entry.

The main input components of the system include student personal information, academic performance details, co-curricular activity details, and mentoring task information. These inputs are stored in the system database and used for analyzing student performance and identifying students who require academic support.

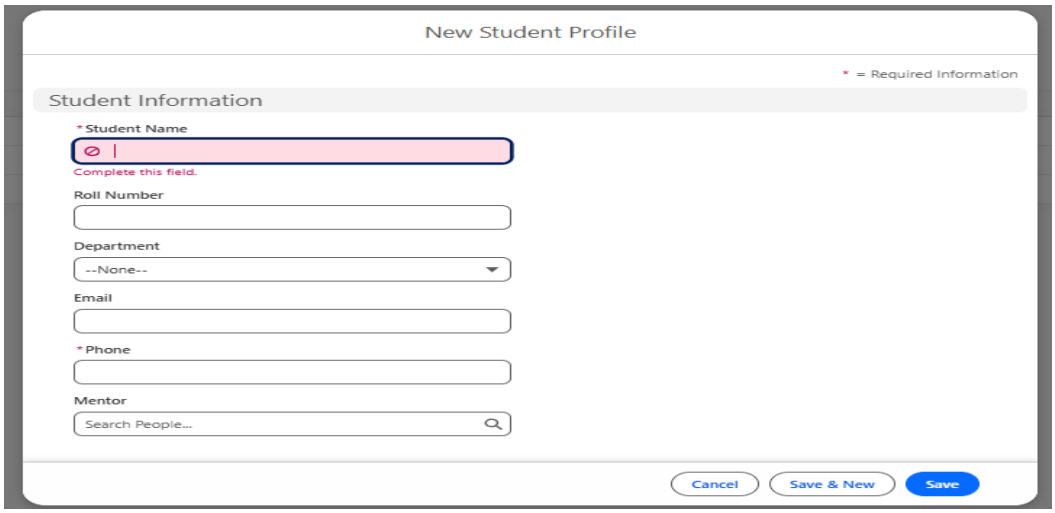

Figure 3.4 Student Personal Detail Entry Form

In Figure3.4, This form allows students to enter their basic personal information. The details entered are stored in the student database and used to identify and manage student records.

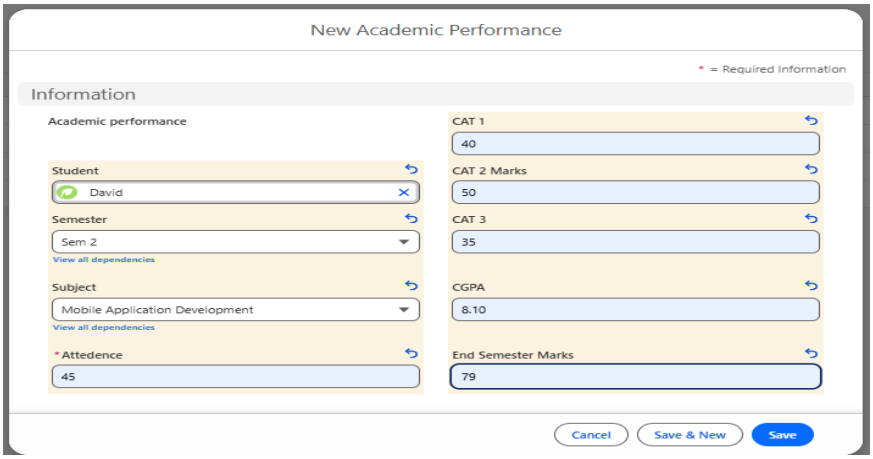

Figure 3.5 Academic Performance Entry Form

In Figure 3.5, This form allows students to enter their academic performance details such as subject marks and attendance. These inputs help the system analyze student performance and detect students with low academic results.

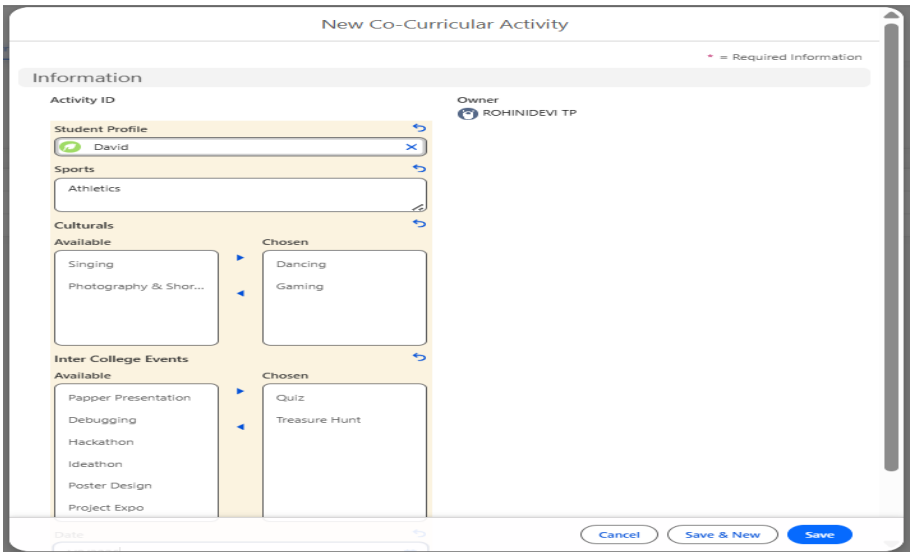

Figure 3.6 Co-Curricular Activities Entry Form

In Figure 3.6, This form allows students to enter details of their co-curricular or extracurricular activities. This information helps mentors understand the student's overall development.

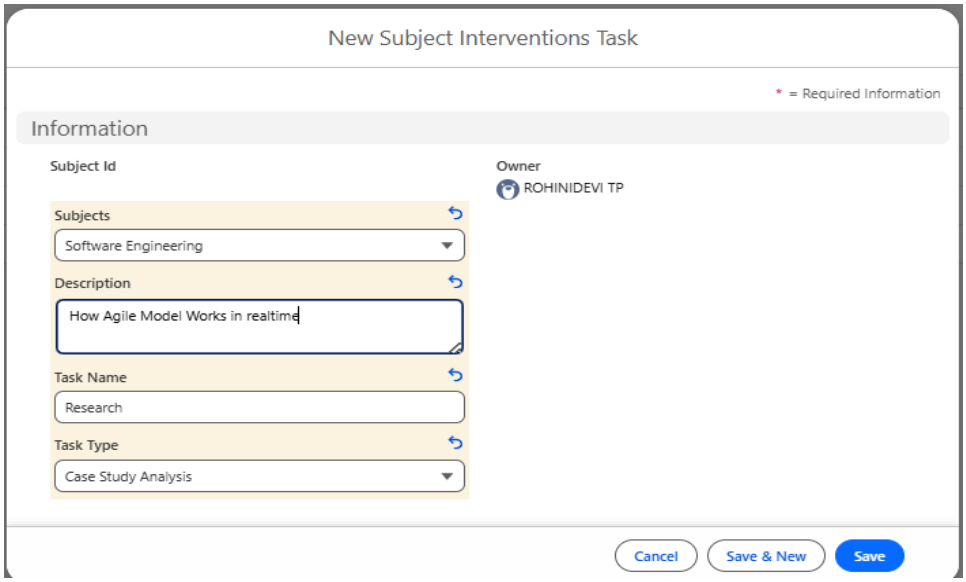

Figure 3.7 Subject Intervention Task Input Form

In Figure 3.7, The information entered through this form is stored in the system database and is later used to notify students about the assigned tasks.

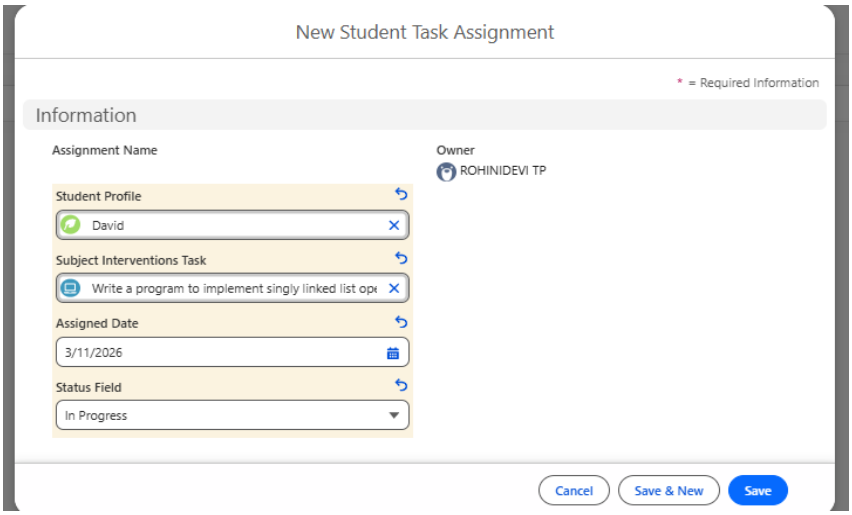

Figure 3.8 Student Task Assignment Input Form

In Figure 3.8, The Student Task Assignment input form helps mentors monitor and guide students who are academically at risk. By assigning subject-specific tasks, mentors can provide structured support to students and track their improvement through the system.

3.5 OUTPUT DESIGN

Output design refers to the process of presenting processed data or information to users in a meaningful and understandable format. The main objective of output design is to provide useful information to users so that they can make decisions or take appropriate actions based on the system results. In the Smart Student Academic Progress Monitoring and Early Risk Detection System, the output design provides information related to student academic performance, mentoring activities, and assigned tasks. The system generates outputs in different forms such as reports, notifications, and task status updates. These outputs help mentors monitor student progress and allow students to track their assigned tasks and academic performance. The system displays output information through a web-based interface developed using the Salesforce platform, making it easy for users to access the results from any location. The major outputs produced by the system include the following.

3.5.1 Student Performance Status

The system analyzes the academic performance of students based on marks and attendance records. The output shows whether the student performance is normal or if the student is academically at risk. This information helps mentors quickly identify students who need academic guidance.

3.5.2 Low Performance Student Alert

When the system detects that a student has low marks or poor attendance, it automatically generates an alert message for the mentor. This output helps mentors take immediate action and provide support to the student. The alert notification is sent through email notifications and also appears within the system interface.

3.5.3 Mentor Assigned Task Notification

After reviewing the student's academic performance, the mentor can assign intervention tasks. The system then generates an output notification informing the student about the assigned task. The notification contains information such as the task name, description, subject, and status of the task.

3.5.4 Student Task Status

Students can view the status of the tasks assigned by the mentor. The output displays whether the task is in progress, completed, or pending. This helps students track their progress and ensures that assigned tasks are completed on time.

3.5.5 Reports and Dashboard

The system also generates reports and dashboard outputs that summarize student academic performance and mentoring activities. These reports help mentors and administrators monitor the overall performance of students and identify those who require academic support. The dashboard presents the information in the form of tables, charts, and summary reports, making it easier to analyze student data.

3.6 ER DIAGRAM

The Entity Relationship Diagram represents the logical structure of the database used in the Smart Student Academic Progress Monitoring and Early Risk Detection System. It shows the relationships between different entities involved in monitoring student academic performance and mentoring activities. The Entity Relationship diagram helps in understanding how data is organized, stored, and connected within the system. It identifies the main entities, their attributes, and the relationships between them.

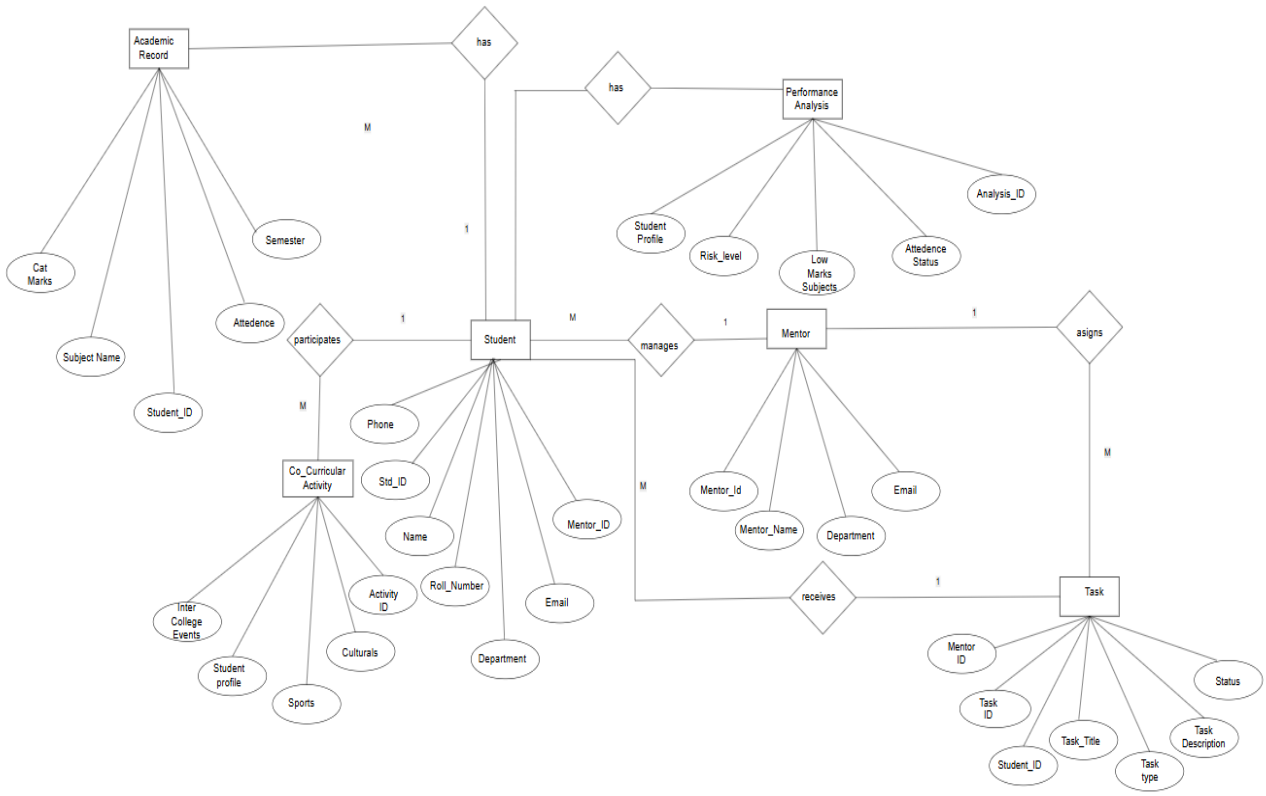

Figure 3.9 Entity Relationship Diagram

The Entity Relationship Diagram represents the structure of the database used in the Smart Student Academic Progress Monitoring and Early Risk Detection System. It illustrates the entities involved in the system, their attributes, and the relationships between them. The ER diagram helps in understanding how student academic data is stored and how mentors interact with the system to monitor and guide students. The main entities involved in the system include Student, Academic Record, Co-Curricular Activity, Performance Analysis, Mentor, and Task.

The Student entity stores the basic details of each student such as Std_ID, Name, Roll_Number, Email, Phone, Department, and Mentor_ID. Each student is assigned to a mentor who monitors their academic progress and provides guidance when necessary. The Academic Record entity stores the academic details of students. It includes attributes such as Student_ID, Subject_Name, CAT Marks, Attendance, and Semester.

A student can have multiple academic records since they may have different subjects and semesters. The Co-Curricular Activity entity records the extracurricular participation of students. It includes attributes such as Activity_ID, Student Profile, Sports, Cultural Activities, and Inter-College Events. A student can participate in multiple activities, and these activities are linked to the student entity. The Performance Analysis entity is responsible for evaluating the academic status of students. It contains attributes such as Analysis_ID, Risk_Level, Low Marks Subjects, and Attendance Status. The system analyzes academic records to determine whether a student is academically at risk. The Mentor entity stores information about mentors who guide students.

It includes attributes such as Mentor_ID, Mentor_Name, Department, and Email. A mentor can manage multiple students and monitor their academic performance. The Task entity represents the intervention tasks assigned by mentors to students. It contains attributes such as Task_ID, Task_Title, Task_Type, Task_Description, Status, Student_ID, and Mentor_ID. These tasks help students improve their academic performance in subjects where they are weak. The Entity Relationship diagram helps in understanding how data is organized, stored, and connected within the system. It identifies the main entities, their attributes, and the relationships between them.

3.7 DATABASE DESIGN

The database design defines how data is organized, stored, and managed in the Smart Student Academic Progress Monitoring and Early Risk Detection System. The system uses the Salesforce Cloud Database, which stores data using objects, fields, and relationships. The database is designed to maintain student academic information, mentoring activities, and intervention tasks in an organized and structured manner.

The main entities used in the system include Student, Academic Record, Co-Curricular Activity, Performance Analysis, Mentor, and Task. These entities are connected through relationships to maintain data integrity and ensure efficient retrieval of information. The database supports the process of monitoring student academic performance, detecting students who are academically at risk, and allowing mentors to assign intervention tasks.

3.8 TABLE DESIGN

Table design describes the structure of each table used in the database. Each table contains fields that store specific information related to students, academic records, mentors, and intervention tasks.

Table 3.1 Student Table

Field Name	Data Type	Description
Std_ID	Auto Number	Unique identification number automatically generated for each student
Name	Text	Stores the name of the student
Roll_Number	Text	Stores the roll number assigned to the student
Email	Email	Stores the email address of the student
Phone	Phone	Stores the contact number of the student
Department	Text	Stores the department in which the student is studying
Mentor_ID	Lookup (Mentor)	Stores the mentor assigned to monitor the student

Above Table 3.1, The Student table stores the personal and academic identification details of students who use the system.

Table 3.2 Academic Record Table

Field Name	Data Type	Description
Academic_Record_ID	Auto Number	Unique identifier for each academic performance record
Student_ID	Lookup (Student)	References the student whose academic details are stored
Subject_Name	Text	Stores the name of the subject
CAT_Marks	Number	Stores the marks obtained by the student
Attendance	Percent	Stores the attendance percentage of the student
Semester	Number	Stores the semester number of the subject

Above Table 3.2, The Academic Record table stores the academic performance details of students.

Table 3.3 Co-Curricular Activity Table

Field Name	Data Type	Description
Activity_ID	Auto Number	Unique identifier for each co-curricular activity record
Student_Profile	Lookup (Student)	References the student participating in the activity
Sports	Text	Stores information about sports participation
Cultural	Text	Stores information about cultural activities
Inter_College_Events	Text	Stores details of inter-college events attended by the student

Above Table 3.3, The Co-Curricular Activity table records the extracurricular activities in which students participate.

Table 3.4 Performance Analysis Table

Field Name	Data Type	Description
Analysis_ID	Auto Number	Unique identifier for each performance analysis record
Student_Profile	Lookup (Student)	References the student whose academic performance is analyzed
Risk_Level	Text	Indicates the academic risk level of the student
Low_Marks_Subjects	Text	Stores the subjects where the student scored low marks
Attendance_Status	Text	Indicates whether the student attendance is satisfactory or low

Above Table 3.4, The Performance Analysis table is responsible for evaluating the academic status of students.

Table 3.5 Mentor Table

Field Name	Data Type	Description
Mentor_ID	Auto Number	Unique identifier for each mentor
Mentor_Name	Text	Stores the name of the mentor
Department	Text	Stores the department of the mentor
Email	Email	Stores the email address of the mentor

Above Table 3.5, The Mentor table stores the details of mentors who guide and monitor students. It includes information such as mentor name, department, and email address.

Table 3.6 Task Table

Field Name	Data Type	Description
Task_ID	Auto Number	Unique identifier for each intervention task
Student_ID	Lookup (Student)	References the student who must complete the task
Mentor_ID	Lookup (Mentor)	References the mentor assigning the task
Task_Title	Text	Stores the title of the assigned task
Task_Type	Picklist	Specifies the type of intervention task
Task_Description	Long Text Area	Provides detailed instructions for the assigned task
Status	Picklist	Indicates whether the task is pending, in progress, or completed

Above Table 3.6, The Task table stores the intervention tasks assigned by mentors to students.

SUMMARY

This chapter described the system design of the Smart Student Academic Progress Monitoring and Early Risk Detection System. It explained the different modules included in the system such as the student module, mentor module, academic performance module, co-curricular activity module, task assignment module, and report module. The chapter also presented the Data Flow Diagrams (DFD), System Flow Diagram, and Entity Relationship (ER) Diagram, which illustrate how data flows through the system and how different entities are related. In addition, the chapter explained the database design and table structures used to store student information, academic records, mentoring tasks, and performance analysis data. The input and output designs were also discussed to show how users interact with the system and how the processed information is presented. Overall, this chapter provides a clear understanding of the system architecture and the design components required for implementing the proposed system effectively.

CHAPTER 4

4. IMPLEMENTATION

4.1 CODE DESCRIPTION

The system is implemented using the Salesforce development environment, which primarily uses configuration-based development instead of traditional programming code. The system logic is implemented through Salesforce Flows, Process Automation, Validation Rules, and Email Alerts. Custom objects are created to store information related to students, academic performance, co-curricular activities, mentor details, and intervention tasks. Relationships between objects are established using lookup relationships, allowing data to be connected across different modules. Automation is implemented using Salesforce Flows, which analyze student academic data and trigger actions when certain conditions are met.

For example, if the system detects low marks or low attendance, an automated flow sends an email alert to the assigned mentor. This automation ensures that mentors are informed quickly and can take appropriate action. Reports and dashboards are also created to provide summarized information about student performance and mentoring activities. These components help mentors monitor academic trends and identify students who require academic guidance.

4.2 STANDARDIZATION OF THE CODING

Standardization of coding ensures that the system is developed using consistent naming conventions, structured design principles, and maintainable components. In this project, Salesforce development standards are followed to ensure that the system is organized and easy to manage. All custom objects, fields, and components are created using meaningful names to clearly represent their purpose within the system. For example, objects such as Student, Academic Record, Co-Curricular Activity, Mentor, and Subject Intervention Task are used to represent specific system components.

Validation rules are implemented to ensure that users enter correct and complete information when submitting data. Field-level security and access controls are used to restrict access to sensitive information, ensuring that only authorized users can view or modify certain records. Following these standard practices improves system reliability, simplifies maintenance, and ensures that the system can be easily updated or expanded in the future.

4.3 ERROR HANDLING

Error handling is an important aspect of system implementation that ensures the system operates smoothly and prevents incorrect data from being stored in the database. In this system, error handling is implemented through validation rules, required fields, and data format checks.Validation rules are used to ensure that users provide valid input when entering information into the system.

For example, required fields must be filled before the record can be saved, and fields such as email addresses and phone numbers must follow proper formats. The system also prevents incorrect academic data from being entered. For instance, marks and attendance values must be within acceptable ranges. If invalid data is entered, the system displays an error message and prompts the user to correct the information before submitting the record. These error handling mechanisms help maintain data accuracy and ensure that the system functions reliably.

SUMMARY

This chapter described the implementation of the Smart Student Academic Progress Monitoring and Early Risk Detection System using the Salesforce platform. It explained how the system components were developed using custom objects, flows, and automation features. The chapter also discussed the coding standards followed during development and the error handling mechanisms used to ensure data accuracy and system reliability. The implementation phase successfully transformed the system design into a functional application that supports monitoring student performance and providing mentoring support

CHAPTER 5
TESTING AND RESULTS

5.1 SYSTEM TESTING

Testing is an important phase in the software development life cycle that ensures the reliability, accuracy, and performance of the Student Progress Monitoring System. This chapter explains the testing methods used to verify that the developed system meets its functional requirements and operates correctly under different conditions. Testing was conducted after the completion of system development, and the results were documented using appropriate test cases.

The system was tested at multiple levels to detect and correct errors, verify module functionality, and ensure smooth interaction between different system components. Each module was first tested individually and later tested as part of the complete system. The testing methods used in this project include:

- Unit Testing
- Integration Testing
- Validation Testing

5.1.1 unit testing

Unit testing focuses on verifying individual modules of the system independently to ensure that each module performs its intended functionality correctly. In this project, unit testing was carried out during the development stage. Each module such as Student Profile, Academic Performance, Co-Curricular Activities, Performance Analysis, Mentor Module, and Task Assignment Module was tested separately.

CASE 1

Module: Student Profile Module

Test Type: Data Entry Verification

Input: Student Name, Roll Number, Department, Email

Output: Student record created successfully

TEST

Input: Name = “Rohini”, Roll No = “24MCR086”, Department = “MCA”

Output: Student record saved and displayed

Analysis: The system correctly stored student details.

Result: Pass

CASE 2

Module: Academic Performance Module

Test Type: Marks Entry Verification

Input: CAT1, CAT2, CAT3, End Semester Marks

Output: Marks stored successfully

TEST

Input: CAT1 = 25, CAT2 = 30, CAT3 = 35, End Sem = 80

Output: Marks saved and displayed

Analysis: Marks are recorded correctly.

Result: Pass

CASE 3

Module: Co-Curricular Activities Module

Test Type: Activity Entry

Input: Activity Name, Type, Date

Output: Activity stored successfully

TEST

Input: Sports = “Cricket”, Date = 12-02-2026

Output: Activity record displayed

Analysis: Activity data stored properly.

Result: Pass

CASE 4

Module: Performance Analysis Module

Test Type: Risk Detection

Input: Marks < 40

Output: Student marked as high-risk

TEST

Input: End Semester Marks = 30

Output: Student flagged as high-risk

Analysis: Risk condition applied correctly.

Result: Pass

CASE 5

Module: Task Assignment Module

Test Type: Task Assignment

Input: Assign task to student

Output: Task assigned successfully

TEST

Input: Task = “Improve DBMS”, Student = Rohini

Output: Task displayed in student record

Analysis: Task linked correctly.

Result: Pass

5.1.2 Integration Testing

Integration testing verifies the interaction between different modules to ensure that data flows correctly between system components. In this project, modules such as Student Profile, Academic Performance, Performance Analysis, Mentor Notification, and Task Assignment were integrated and tested together. This testing ensures that the outputs of one module are correctly used as inputs for another module. It also confirms that the system operates smoothly when multiple features are used together.

CASE 1

Module: Student + Academic Module

Test Type: Data Integration

Input: Enter marks for student

Output: Academic data linked to student

TEST

Input: Student = Rohini, Marks entered

Output: Marks displayed under student

Analysis: Data linked successfully between modules.

Result: Pass

CASE 2

Module: Academic + Performance Analysis

Test Type: Data Processing

Input: Low marks

Output: Student identified as high-risk

TEST

Input: Marks = 35

Output: Risk status updated

Analysis: Integration works correctly.

Result: Pass

CASE 3

Module: Performance + Mentor Module

Test Type: Alert System

Input: High-risk student

Output: Mentor receives alert

TEST

Input: Student marked as high-risk

Output: Notification sent to mentor

Analysis: Alert system functioning properly.

Result: Pass

CASE 4

Module: Mentor + Task Assignment

Test Type: Task Flow

Input: Assign task

Output: Task visible to student

TEST

Input: Mentor assigns task

Output: Student receives task

Analysis: Task flow is correct.

Result: Pass

5.1.3 Validation Testing

Validation testing ensures that the system meets user requirements and produces correct results. It confirms that the system functions according to the intended design and provides accurate outputs. In this project, validation testing verifies that the system correctly identifies students with low performance and provides proper mentoring support.

CASE 1

Module: Student Module

Test Type: Required Field Validation

Input: Empty Name field

Output: Error message displayed

TEST

Input: Name = (empty)

Output: “Field required” error

Analysis: Validation works correctly.

Result: Pass

CASE 2

Module: Academic Module

Test Type: Input Validation

Input: Marks > 100

Output: Error message

TEST

Input: Marks = 120

Output: Invalid input error

Analysis: System prevents incorrect data entry.

Result: Pass

SUMMARY

This chapter discussed the testing procedures used to evaluate the Smart Student Academic Progress Monitoring and Early Risk Detection System. Various testing methods such as unit testing, integration testing, and validation testing were conducted to ensure that the system operates correctly. The testing process verified that student data entry, academic performance analysis, mentor alerts, and task assignment functionalities work effectively. The successful testing results confirm that the system functions reliably and meets the intended project requirements.

CHAPTER – 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 CONCLUSION

The Student Progress Monitoring System was successfully developed to manage and analyze student academic performance. The system helps in storing student details, tracking marks and attendance, and identifying high-risk students. It also enables mentors to take necessary actions through alerts and task assignments. Overall, the system improves academic monitoring and supports better decision-making.

6.2 FUTURE ENHANCEMENTS

Although the system meets current requirements, several enhancements can be implemented to improve its functionality and performance in the future.

- Mobile Application Integration
  A mobile version of the system can be developed to allow students and mentors to access data anytime and anywhere.
- AI-Based Performance Prediction
  Machine learning algorithms can be integrated to predict student performance and identify high-risk students at an early stage.
- Parent Notification System
  The system can be extended to send notifications to parents regarding student performance and attendance.
- Advanced Analytics Dashboard
  More advanced dashboards can be added to provide deeper insights such as subject-wise analysis and trend prediction.
- Integration with Learning Platforms
  The system can be integrated with Learning Management Systems (LMS) to automatically fetch assignment and exam data.
- Automated Recommendation System
  The system can suggest personalized learning materials and improvement strategies for students.

APPENDICES

A. SAMPLE CODING


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</CustomField>
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    <type>Number</type>
    <unique>false</unique>
</CustomField>
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B. SCREENSHOTS

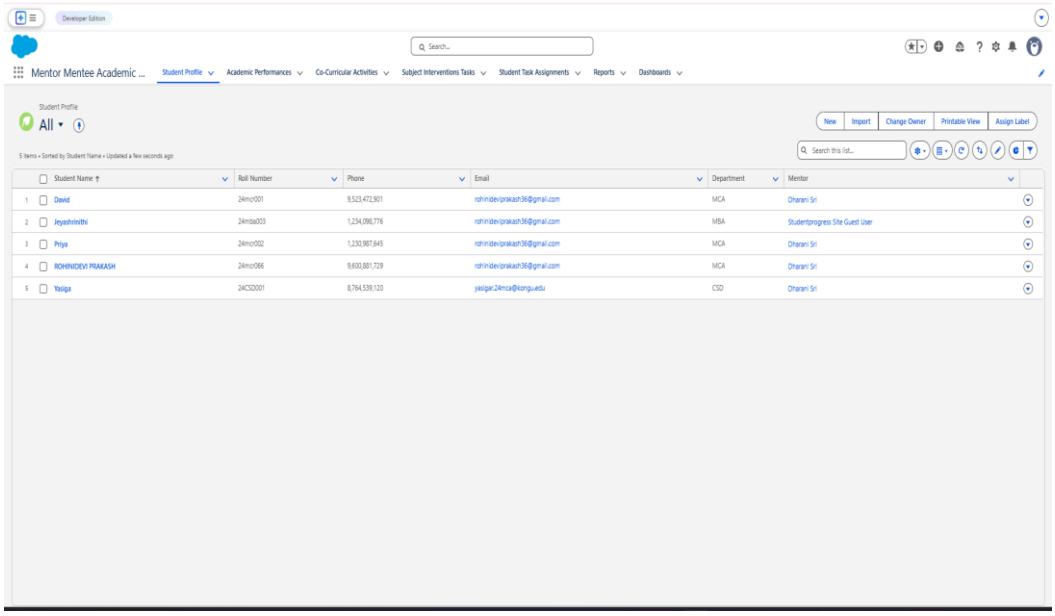

Figure B.1 Student Profile List View

In Figure B.1, This screen displays the Student Profile list view, showing details such as student name, roll number, email, department, and mentor. It helps faculty and mentors to easily manage and access student information in a centralized system.

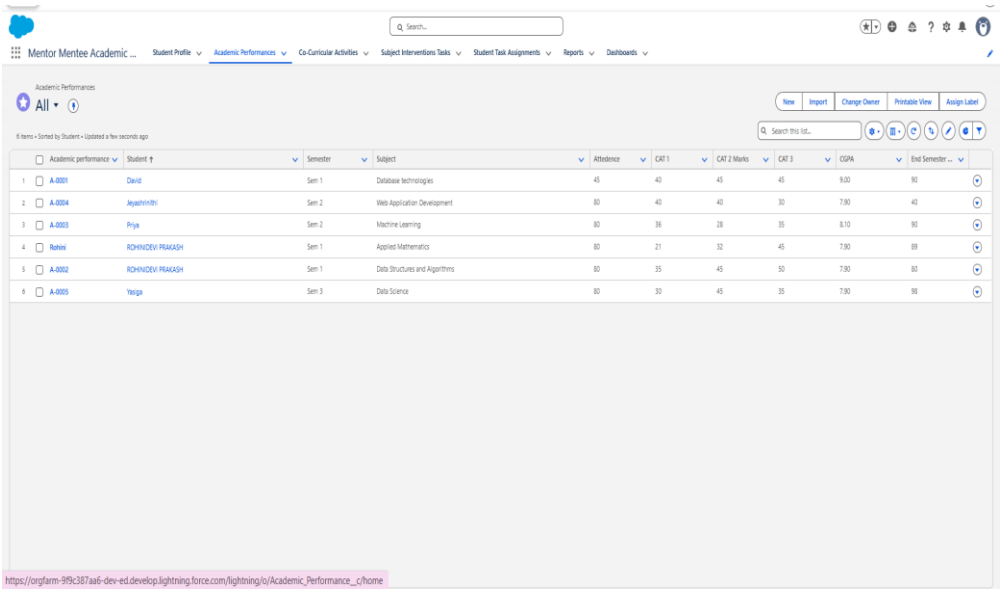

Figure B.2 Academic Performance List View

In Figure B.2, This screen displays the Academic Performance list view, showing student marks, attendance, CGPA, and subject details. It helps faculty to analyze student performance and identify students who require academic support.

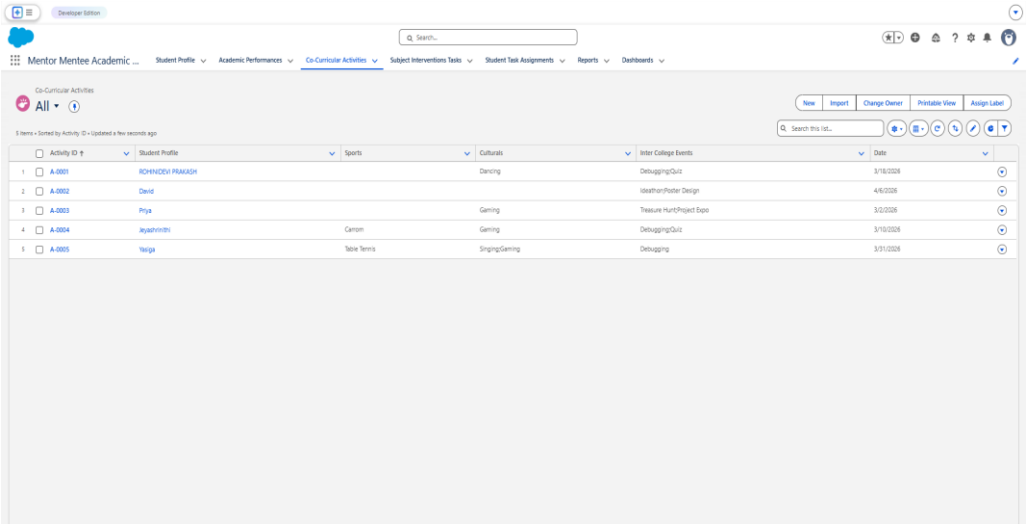

Figure B.3 Co-Curricular Activities List View

In Figure B.3, This screen displays the Co-Curricular Activities list view, showing student participation in sports, cultural activities, and inter-college events. It helps mentors track student engagement beyond academics and evaluate overall development.

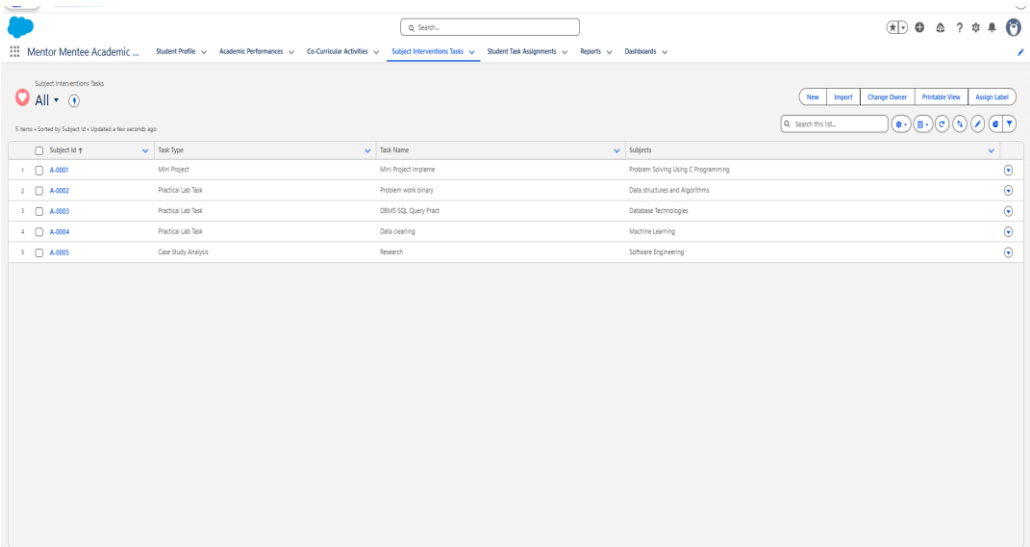

Figure B.4 Subject Intervention Tasks List View

In Figure B.4, This screen displays the Subject Intervention Tasks list view, showing tasks assigned for different subjects to improve student performance. It helps mentors plan and manage academic interventions effectively based on student needs.

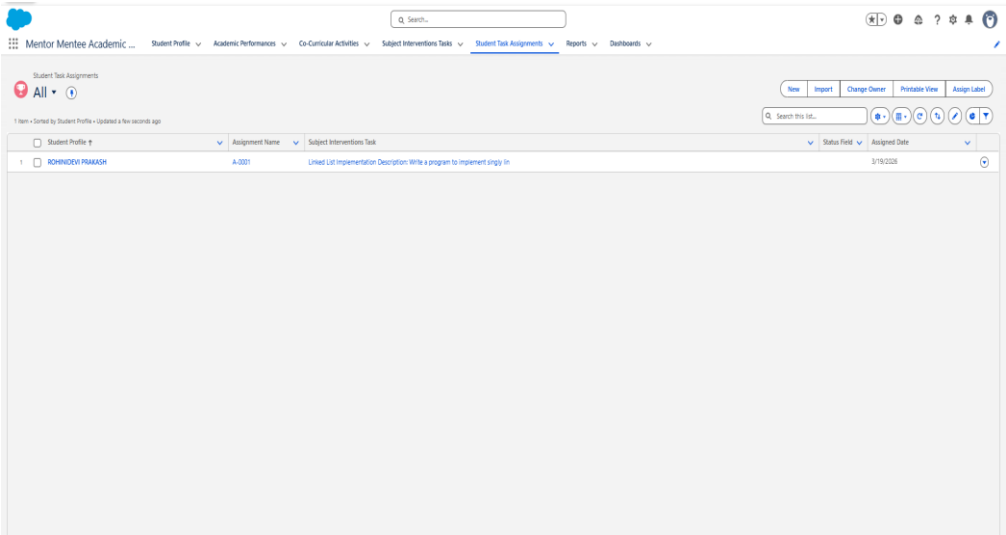

Figure B.5 Student Task Assignment List View

In Figure B.5, This screen displays the Student Task Assignment list view, showing tasks assigned to students along with status and assigned date. It helps mentors track task completion and monitor student progress effectively.

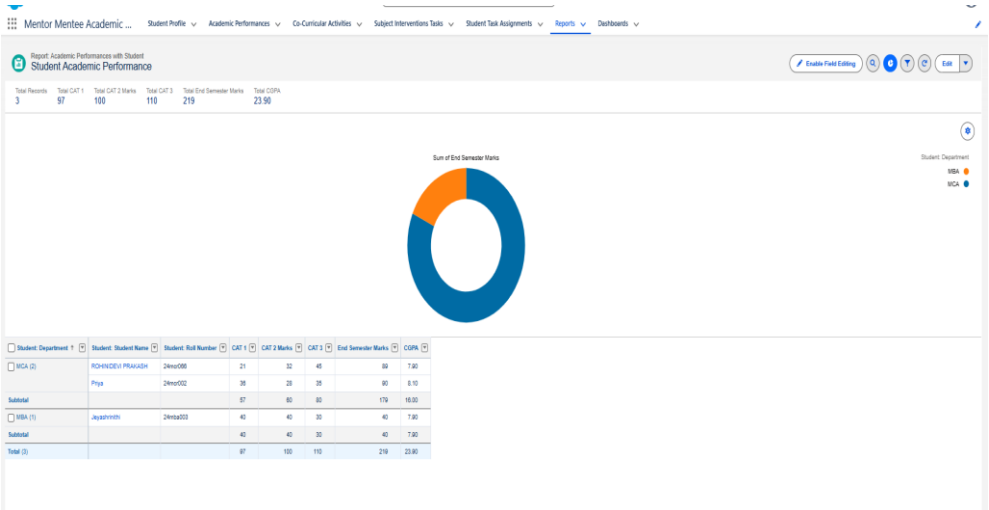

Figure B.6 Student Academic Performance Report

In Figure B.6, This screen displays the Student Academic Performance Report with graphical visualization of marks and department-wise analysis. It helps in identifying performance trends and detecting students who require academic support.

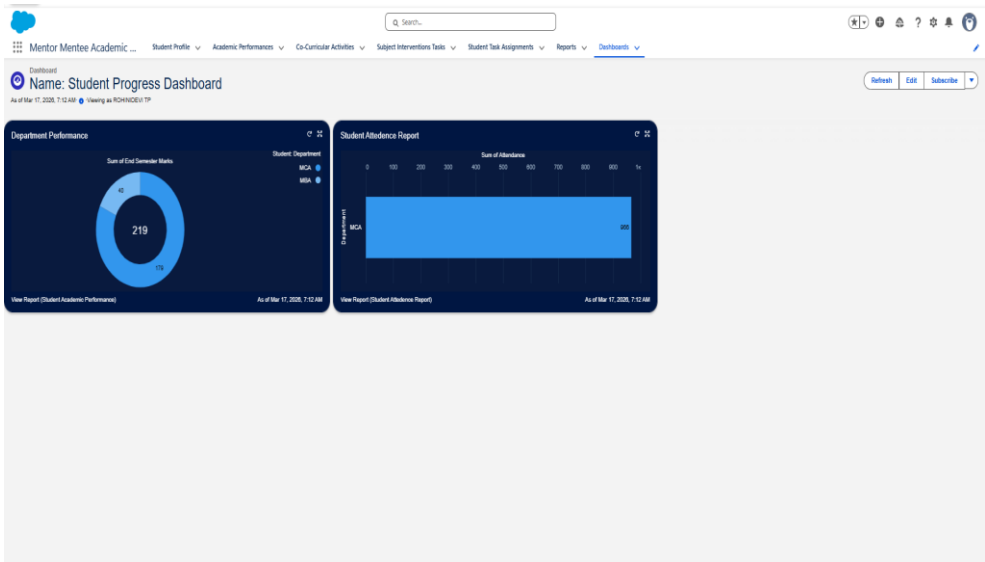

Figure B.7 Student Progress Dashboard

In Figure B.7, This screen displays the Student Progress Dashboard with visual charts representing academic performance and attendance. It helps mentors quickly analyze student data and identify areas that require attention.

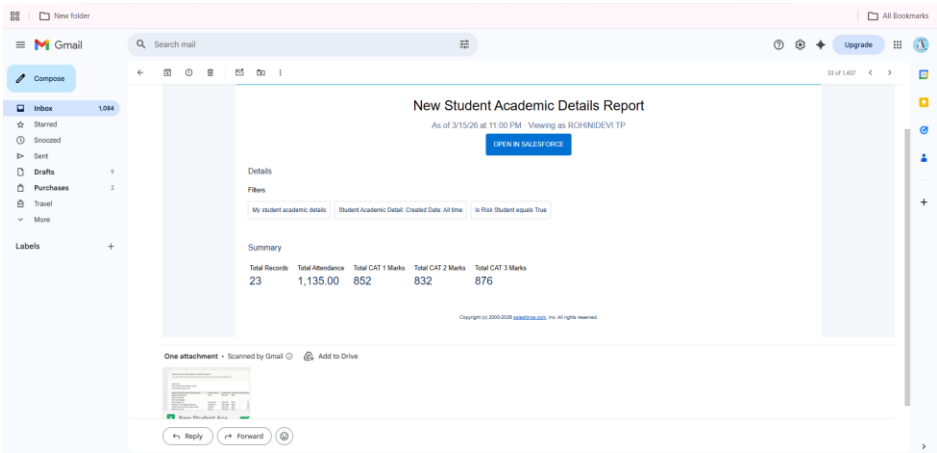

Figure B.8 Student Academic Details Email Report

In Figure B.8, This screen displays the automated email report generated by the system, showing student academic details and performance summary.

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MAPPING OF PROJECT GOALS TO SDG

SUSTAINABLE DEVELOPMENT GOALS (SDGs)

SDG 4 – Quality Education

This goal focuses on ensuring inclusive and quality education for all and promoting lifelong learning opportunities. The **Student Progress Monitoring System** supports this goal by providing a digital platform to track and analyze student academic performance. It helps in identifying students who need academic support and enables mentors to guide them effectively, thereby improving learning outcomes and overall educational quality.

SDG 9 – Industry, Innovation and Infrastructure

This goal promotes innovation and the use of technology in developing efficient systems. The proposed system supports this goal by introducing a digital solution for managing student data, academic records, and performance analysis. It enhances institutional infrastructure by replacing manual processes with an automated and scalable system.

SDG 8 – Decent Work and Economic Growth

This goal emphasizes improving productivity and efficiency through technology. The system helps educational institutions streamline academic monitoring, reduce manual workload, and improve decision-making. By improving student performance and skill development, it indirectly contributes to better career opportunities and economic growth.

SDG 12 – Responsible Consumption and Production

The system promotes efficient use of academic resources by providing accurate data analysis and reports. It reduces unnecessary manual work and paper usage by digitizing student records and reports, thereby supporting sustainable and responsible resource management.